EPSR-test-ex35 randomise (+ mixato, fmole)

Program(s): mixato, randomise, fmole | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises `randomise` (subroutine of `introtcluster.f`), which gives every molecule of a box a random position and a random whole-molecule orientation using the random-number state stored in the `.ato` file — the step that turns the stacked molecules written by `mixato` into a disordered starting configuration for EPSR [1,2].

Code exercised

Fortran (original, fortran_src/):

- `introtcluster.f`: `randomise` (random positions if `stepm` > `0`, random rotations if `stepr` > `0`)
- `molfit.f`: `rotatm` (whole-molecule rotation); `auxiliary_routines.f`: `ran1`
- `ato_read_and_save.f`: `minitc` (read `.ato`), `moloutc` (write `.ato`), `normalise_coordinates`; `minimagec.f`: `minimagec`
- `mixato.f`, `fmole.f` (box building and shaking in between); `file_and_directory_commands.f`: `search_for_files` (file prompt / directory listing), `get_full_filename`

Rust (port, rust/epsr-core/src/):

- `tools/randomise.rs`: `run`
- `tools/setup_common.rs`: `rotatm`, `search_for_files`; `rng.rs`: `Ran1::ran1`
- `ato.rs`: `minitc`, `moloutc`, `normalise_coordinates`; `neighbour.rs`: `minimagec`
- `tools/mixato.rs`, `tools/fmole.rs`

Physical system

Dimethyl sulphoxide-water mixtures [3] at 300 K: a new box of 15 DMSO + 30 water (240 atoms) at 0.09 atoms/Å^3, cubic side 13.867 Å (`box.ato`), and the full EPSR-test configuration of 200 DMSO + 400 water (3200 atoms) in a cubic box of 33.1213 Å. Random numbers come from the Park-Miller generator with Bays-Durham shuffle (`ran1`) [4,5], whose state is saved in every `.ato` file.

Inputs

`DMSO.ato`, `WaterMolecule.ato`, `200DMSO400Water.ato` from EPSR-test; `mixato.stdin` (15 DMSO, 30 water, density 0.09, output `box`); `randomise.stdin` (file name `box`). Commands: `mixato`, `randomise box`, `fmole box 20`, `randomise < randomise.stdin` (name typed), `randomise 200DMSO400Water.ato`.

How the accuracy was verified

`commands.txt` (5 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

Five program calls in one pass, each reading the file written by the previous one, so any difference in the random-number sequence would propagate. Compared outputs: `box.ato`, `200DMSO400Water.ato` and `dirlist.txt`, all identical. The randomisation is visible in the reference: the first molecule of `box.ato` sits at (-6.03, 4.65, 6.55) Å instead of the template position near the origin written by `mixato`.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).

[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).

[3] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).

[4] S. K. Park and K. W. Miller, "Random number generators: good ones are hard to find", Commun. ACM 31, 1192-1201 (1988).

[5] W. H. Press, S. A. Teukolsky, W. T. Vetterling and B. P. Flannery, Numerical Recipes in Fortran 77, 2nd ed., Cambridge University Press (1992).

[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.